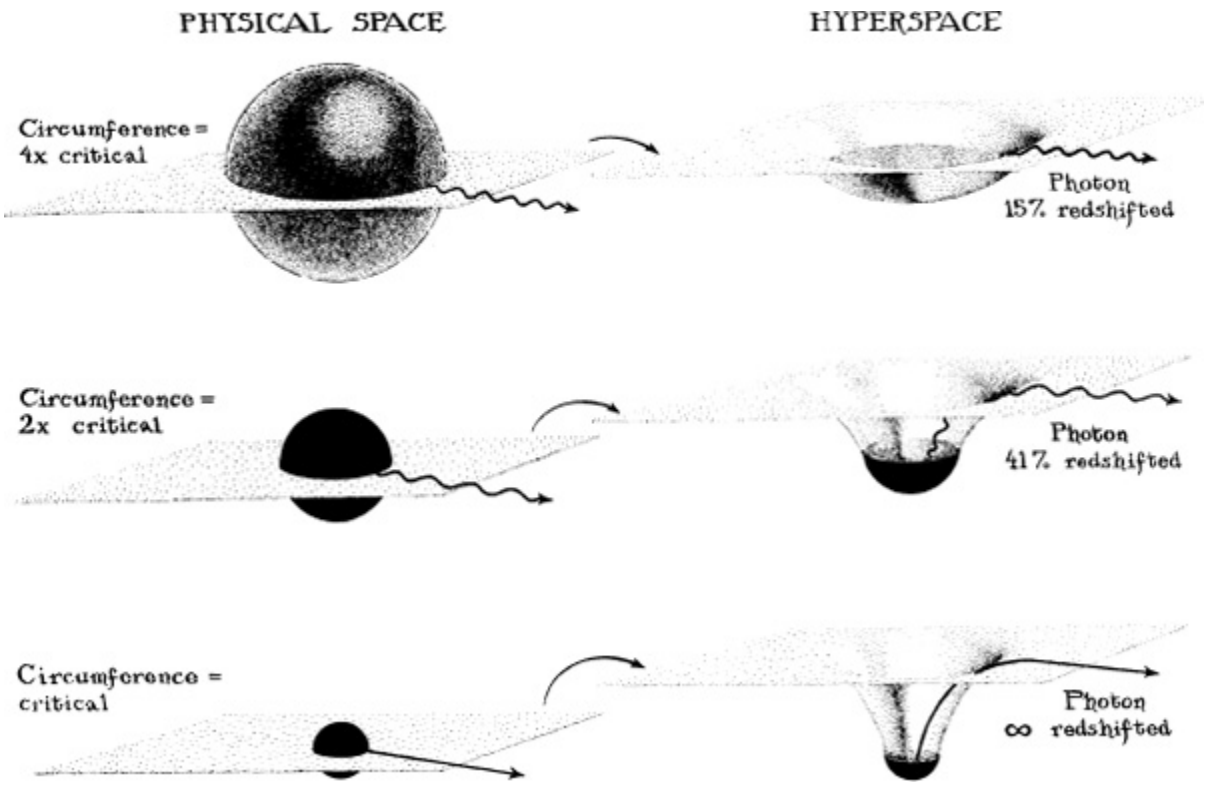
be redshifted by 41 percent. These predictions seemed acceptable and reasonable. What did not seem at all reasonable to physicists and astrophysicists of the 1920s, or even as late as the 1960s, was the prediction for a star whose actual circumference was the same as its critical one (bottom part of Figure 3.4). For such a star, with its more strongly curved space, the flow of time at the star's surface is infinitely dilated; time does not flow at all—it is frozen. And correspondingly, no matter what may be the color of light when it begins its journey upward from the star's surface, it must get shifted beyond the red, beyond the infrared, beyond. radio wavelengths, all the way to infinite wavelengths; that is, all the way out of existence. In modern language, the star's surface, with its critical circumference, is precisely at the horizon of a black hole; the star, by its strong gravity, is creating a black-hole horizon around itself.



3.4 General relativity's predictions for the curvature of space and the redshift of light from three highly compact stars with the same mass but different circumferences. The first is four times larger than the critical circumference, the second is twice as large as critical, and the third has its circumference precisely critical. In modern language, the surface of the third star is a black-hole horizon.

The bottom line of this Schwarzschild-geometry discussion is the same as that found by Michell and Laplace: A star as small as the critical